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Genetic variation at rs11574891 and rs3125003 in NOTCH1 is associated with therapeutic efficacy of Janus kinase inhibitors in Taiwanese patients with rheumatoid arthritis

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ABSTRACT

Objectives: Janus kinase inhibitors (JAKi) have been widely adopted in the treatment of rheumatoid arthritis (RA); however, inter-individual variability in therapeutic response remains a significant challenge. Pharmacogenomic data specific to the Taiwanese population are still limited. This study aimed to identify single nucleotide polymorphisms (SNPs) associated with JAKi efficacy in Taiwanese RA patients.

Methods: Using data from the Taiwan Precision Medicine Initiative, we analyzed RA patients receiving JAKi (n = 34) and non-JAKi treatment (n = 376) through a genome-wide association study. In a subset of patients, pre-treatment peripheral blood samples were collected from 15 JAKi-treated and 54 non-JAKi individuals for immunophenotyping and polygenic risk score (PRS) modeling.

Results: GWAS identified 26 SNPs significantly associated with JAKi treatment response. Four SNPs—rs11574891, rs3125003, rs61807669, and Affx-4817039—showed significant associations with B cell subsets previously shown to predict JAKi response. Among them, rs11574891 (OR = 3.67) and rs3125003 (OR = 3.14), both located within *NOTCH1* gene, were independently validated using clumping-and-thresholding and PRSice-2 models. Furthermore, rs75547179 in the *CSMD2* gene was consistently identified across four independent PRS models.

Conclusions: To our knowledge, this is the first pharmacogenomic study in a Taiwanese RA cohort to identify these SNPs and propose rs11574891 and rs3125003 in *NOTCH1* as potential predictive biomarkers for JAKi efficacy. Further validation in larger, independent cohorts is warranted to facilitate the implementation of precision medicine in RA treatment.

1. Introduction

Rheumatoid arthritis (RA) is a chronic, systemic autoimmune disease with a global prevalence of approximately 1 %, and it exhibits a significantly higher incidence in women than in men [1,2]. Clinically, RA is characterized by persistent synovial inflammation and progressive joint destruction, ultimately leading to functional disability in approximately 15 % of patients [1]. Conventional synthetic disease-modifying antirheumatic drugs (csDMARDs) represent the first-line therapy; however, for patients with moderate-to-severe disease who demonstrate an inadequate response to csDMARDs, current treatment guidelines recommend escalation to biologic DMARDs (bDMARDs) or targeted synthetic DMARDs (tsDMARDs) [3].

Janus kinase inhibitors (JAKi), a class of orally administered small-molecule tsDMARDs, offer advantages in dosing convenience and patient adherence compared to injectable bDMARDs. By inhibiting the Janus kinase–signal transducer and activator of transcription (JAK–STAT) signaling pathway, JAKi modulates immune responses to achieve therapeutic benefit [1]. Since 2012, multiple JAKi agents—including tofacitinib, baricitinib, peficitinib, and upadacitinib—have received

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approval from the U.S. Food and Drug Administration for the treatment of RA [4]. Beyond their demonstrated efficacy in refractory RA [5,6], JAKi have been used in other autoimmune disorders, certain malignancies [7], and even COVID-19–related pneumonia [4,8].

Single nucleotide polymorphisms (SNPs) are key genetic determinants underlying interindividual variability in disease susceptibility and drug response [9,10]. It has been estimated that approximately 15–30 % of the variability in RA treatment response can be attributed to genetic background [2,11]. Although previous studies have identified SNPs associated with the efficacy of csDMARDs and bDMARDs [11–14], pharmacogenomic research specific to JAKi response remains limited, particularly in the Taiwanese population [15]. Our research team has previously identified 18 immune cell subsets with the potential to predict JAKi treatment response [16]. Integrating JAKi response–related SNPs with these immune cell phenotypes may enhance predictive accuracy, optimize therapeutic strategies, reduce healthcare costs, and prevent irreversible joint damage [8].

Therefore, this study aims to conduct a genome-wide association study (GWAS) to identify genetic variants associated with JAKi efficacy in Taiwanese RA patients and to analyze their associations with immune cell phenotypes predictive of JAKi response. Ultimately, we seek to establish genetic biomarkers for personalized RA therapy and advance precision medicine in this field.

2. Material and methods

2.1. Ethics approval

This study was conducted in accordance with the ethical principles outlined in the 2013 Declaration of Helsinki and was approved by the Institutional Review Board (IRB) of Tri-Service General Hospital (TSGH), Taiwan (Approval Numbers: 1–107–05–040, B202005140, 2–105–05–038). Written informed consent was obtained from all participants.

2.2. Study participants and genotyping

Participants were recruited from TSGH as part of the Taiwan Precision Medicine Initiative (TPMI), a collaborative effort between Academia Sinica and 15 leading medical centers in Taiwan. TPMI aims to establish a comprehensive clinical and genetic database comprising one million participants.

Each participant provided approximately 5 mL of peripheral blood, from which genomic DNA was extracted from mononuclear cells. Genotyping was performed using the TPMI-specific SNP array, which includes approximately 130,000 known risk variants, 580000 mapping SNPs, and 20,000 copy number variant markers, based on data from the Taiwan Biobank.

The TPMI project specifically recruited individuals of Taiwanese ancestry (n = 23,595,274), including 95 % of Southern Han Chinese origin (n = 22,415,510) and 2.3 % of Indigenous descent (n = 542,691) [17]. Eligible RA patients were identified from the Division of Rheumatology, Immunology, and Allergy at TSGH based on the ICD-10 code M05 and established diagnostic criteria for RA. Patients receiving JAKi therapy—specifically tofacitinib, upadacitinib, or baricitinib—were included.

A total of 410 RA patients were included, comprising 34 in the JAKi group and 376 in the non-JAKi group (control group) (Fig. 1). Among them, 69 patients with pre-treatment immunophenotypic data (15 from the JAKi group and 54 from the control group) underwent further analysis. Patient characteristics are detailed in Table 1. Details of prior

Table 1
Baseline characteristics of subjects in the study.

	Non-JAKi (N = 376)	JAKi (N = 34)	P-value[a]
Age (years)			
Mean (SD)	59.8 (13.1)	61.9 (11.3)	0.249
Median [Min, Max]	61.0 [20.0, 91.0]	63.0 [30.0, 83.0]	
Gender			
Female	304 (80.8 %)	30 (88.2 %)	0.229
Male	72 (19.2 %)	4 (11.8 %)	

JAKi, Janus kinase inhibitors; N, number; SD, standard deviation; Min, minimum; Max, maximum.

[a] The P-values for age were analyzed using t-test, while the P-values for gender were analyzed using chi-square test.

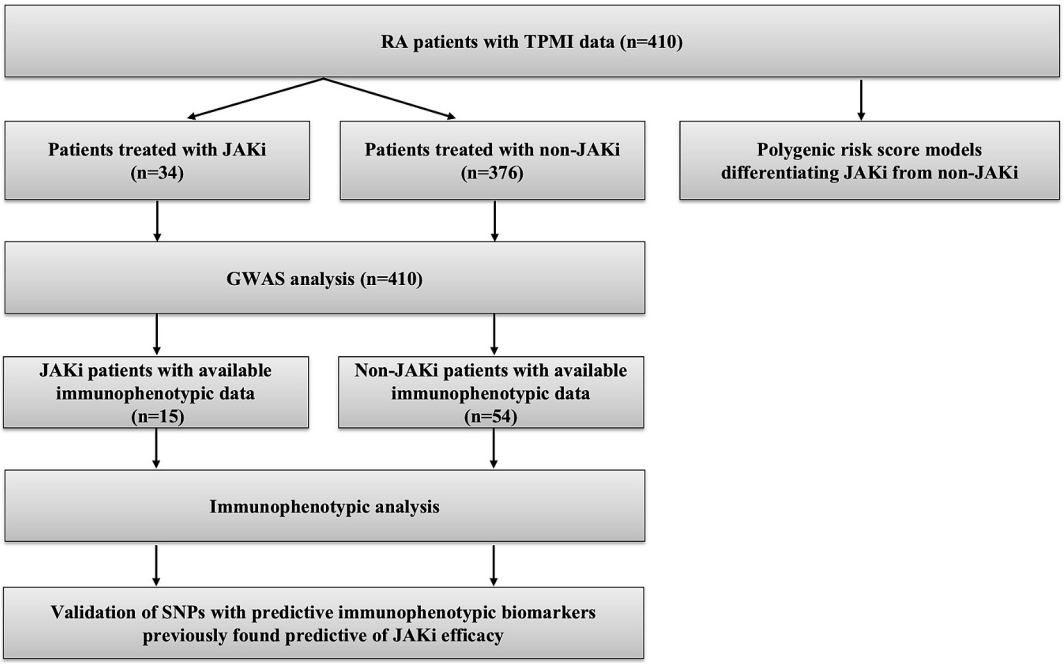


Fig. 1. Flow chart for patient's inclusion and exclusion criteria. RA, rheumatoid arthritis; TPMI, Taiwan Precision Medicine Initiative; JAKi, Janus kinase inhibitors; GWAS, genome-wide association study; SNP, single nucleotide polymorphism.

and current RA medications in the 34 JAKi-treated patients are provided in Supplementary Data 1.

2.3. Association analysis

Quality control and association analysis were conducted using PLINK 1.9, a toolset designed for whole-genome association studies and population-based linkage analyses [18]. Automated quality control procedures, following the pipeline for genetic case-control association studies [19], were implemented. Participants with a genotype missing rate > 0.05 were excluded, and related individuals were removed based on heterozygosity rates and identity-by-descent values. Sex and age were adjusted using regression analysis.

SNPs with call rates <0.95, Hardy-Weinberg equilibrium *P*-values $<1 \times 10^{-4}$, or minor allele frequency < 0.05 were excluded. SNPs with P-values $<1 \times 10^{-4}$ in association analysis were considered statistically significant. Manhattan and Q-Q plots were generated using the R package qqman. Logistic regression analysis with PLINK 1.9 adjusted for sex and age, and odds ratios (OR) with 95 % confidence intervals were calculated to assess SNP associations with JAKi response.

2.4. Functional genomics study of JAKi-associated SNPs

To investigate the functional roles and pathogenic mechanisms of the identified risk SNPs, their genomic locations and associated genes were annotated using the R package ieugwasr (version 0.1.5) (https://mrcieu.github.io/ieugwasr). Functional annotations of the SNPs were derived from several established resources, including the International Genome Sample Resource (https://www.internationalgenome.org), the National Center for Biotechnology Information (https://www.ncbi.nlm.nih.gov), and GeneCards (https://www.genecards.org).

2.5. Blood sample processing and staining for flow cytometry

Blood samples were collected from patients with pre-treatment immunophenotypic data, including 15 patients undergoing JAKi treatment without complications and 54 patients receiving non-JAKi treatment (controls). Blood processing followed standard fluorescence preparation protocols using fluorescent antibody kits with dry reagents (Beckman Coulter, USA).

For the T cell panel (IM T DuraClone cell tube 25 tests; B53328–25 tests, RUO), dry reagents were added with wet antibodies, including Brilliant Violet 605 anti-human CD95 (Fas) antibodies (Catalog number: 305628; BioLegend, San Diego, CA, USA) and Brilliant Violet 650 anti-human CD366 (Tim-3) antibodies (Catalog number: 345028; BioLegend, San Diego, CA, USA). For the regulatory T (Treg) cell panel (IM Treg DuraClone Tube 25 tests; B53346–25 tests, RUO), dry reagents were added with the wet antibodies, including Brilliant Violet 650 anti-human HLA-DR antibodies (Catalog number: 307650; BioLegend, San Diego, CA, USA), Brilliant Violet 605 anti-human CD62L antibodies (Catalog number: 304834; BioLegend, San Diego, CA, USA), Brilliant Violet 785 anti-human CD127 (IL-7Rα) antibodies (Catalog number: 351330; BioLegend, San Diego, CA, USA), and CD8-APC-Alexa Fluor 700 (Catalog number: A66332; Beckman Coulter, USA). For the B cell panel (IM B DuraClone cell tube, 25 tests; B53318–25 tests, RUO), dry reagents were added with wet antibodies, including Brilliant Violet 605 anti-human CD95 (Fas) antibodies (Catalog number: 305628; BioLegend, San Diego, CA, USA) and Brilliant Violet 650 anti-human HLA-DR antibodies (Catalog number: 307650; BioLegend, San Diego, CA, USA). The corresponding fluorescent antibodies are listed in Supplementary Data 2.

After the addition of antibodies, the samples were incubated with the respective reagents. Marker expression levels were then assessed using a 3-laser 13-color CytoFLEX flow cytometer (Beckman Coulter, USA) within 5 h of staining. This was done to detect antibody-fluorescent antigen complexes on the surface of immune cells, enabling differentiation among lymphocyte subgroups and biomarkers.

Flow cytometry standard files were manually processed using CytoFLEX flow cytometry software (Beckman Coulter, USA) to isolate single, live, $CD45^+$ cells. Singlets were gated using forward scatter-area (FSC-A) and forward scatter-width (FSC-W). Flow cytometry data were further analyzed with Kaluza Analysis Software version 2.1 (Beckman Coulter, USA), and specific populations of interest were manually gated. Details of the gating strategy can be found in Supplementary Data 3 to 8. The phenotypes of all cell subsets are listed in Supplementary Data 9.

2.6. Polygenic risk score prediction

Polygenic risk scores (PRS) were generated based on genotype data from RA patients eligible for JAKi treatment and control cohorts, in accordance with established guidelines [20]. Five PRS methods were employed: PRSice2 [21], Clumping and Thresholding (C + T) [22], PRScs (polygenic risk scores–continuous shrinkage) [23], Lassosum [23] and GenEpi [24], with each method implemented according to recommended protocols and default settings. To account for potential confounders, age and gender were included as covariates in all five PRS models.

A train-test split approach was applied, where 10 % of the data was randomly selected as an independent testing dataset to ensure unbiased evaluation of prediction performance. The remaining 90 % was subjected to quality control and GWAS analysis using PLINK, which served as the basis for PRS model construction. All PRS models, except for GenEpi, utilized the same GWAS summary statistics as input but applied different SNP selection algorithms, adjusting inclusion criteria accordingly.

For each method, PRS models were constructed with and without age and gender as covariates to enable direct comparison of predictive performance. The area under the receiver operating characteristic (AUROC) curve was calculated for each model to assess its ability to differentiate JAKi-treated patients from controls. Additionally, the OR for each PRS decile relative to the remaining dataset in the test cohort was computed.

2.7. Statistical analysis

Data preparation, processing, and analysis were conducted using R software (version 4.2.1, R Foundation for Statistical Computing, Vienna, Austria). Categorical variables were expressed as frequencies or percentages, while continuous variables were summarized by mean values with standard deviations or medians with minimum and maximum values. Differences in variables between groups were assessed using the Mann-Whitney *U* test, chi-square test, or *t*-test, as appropriate. OR with 95 % confidence intervals were computed using logistic regression analysis. *P*-values of less than 0.05 were considered statistically significant.

2.8. Post-hoc Power Analysis of SNP Associations with JAKi treatment response

To evaluate the statistical power of detecting SNP associations under different minor allele frequencies (MAF), we employed the R package "genpwr" and applied the genpwr.calc function. A logistic regression model was specified (model = "logistic") to accommodate the binary phenotype. The total sample size was set to $N = 410$ with a case proportion of 8 % (Case.Rate = 0.08). We explored a range of MAF values from 0.07 to 0.34 in increments of 0.01 (MAF = seq (0.07, 0.34, 0.01)), assuming an OR of 3 and 4 at a significance level of $\alpha = 0.05$. Genetic inheritance patterns were modeled under dominant, recessive, and additive frameworks (True.Model = c("Dominant", "Recessive", "Additive")), while the association tests were performed across four genetic models: dominant, recessive, additive, and a 2-degrees-of-freedom genotypic test (Test.Model = c("Dominant", "Recessive", "Additive",

"2df")). This analysis produced a matrix of power estimates for each combination of MAF, true genetic model, and test model. The results were used to assess whether the available sample size provided adequate statistical power to detect SNP–phenotype associations under the assumed genetic effect size and allele frequency distribution.

3. Results

3.1. Identification of 26 SNPs as potential markers for JAKi treatment response

A GWAS was conducted on a cohort of 410 RA patients, consisting of 376 individuals treated with non-JAKi biologics and 34 treated with JAKi. The quality of the analysis was confirmed by a Q-Q plot, which revealed a lambda value of 1.0073 (Fig. 2A). A significance threshold of $P = 1e\text{-}4$ was selected to identify the SNPs with significant associations (Fig. 2B). The annotated names correspond to the SNPs overlapping across five distinct analyses. These analyses included Fisher's exact test, kinship analysis, age-adjusted analysis, sex-adjusted analysis, and analysis adjusted for both age and sex, aimed at accurately identifying SNPs associated with JAKi treatment efficacy (Fig. 2C). The intersection of these approaches revealed 26 SNPs associated with JAKi treatment response (Table 2). Notably, rs161826 on chromosome 1 was associated with the *TNFRSF9* (4-1BB) gene, rs75547179 on chromosome 1 was linked to *CSMD2*, and rs11574891 and rs3125003 on chromosome 9 were mapped to the *NOTCH1* gene. The OR for these SNPs across the different analysis methods were visualized in a heatmap (Fig. 2D), underscoring their potential role in modulating JAKi treatment efficacy.

3.2. The effectiveness of SNP as a predictor of JAKi clinical efficacy

To investigate the correlation between the 26 SNPs and the clinical efficacy of JAKi, we selected patients with pre-treatment immunophenotyping data, comprising 54 individuals treated with non-JAKi biologics and 15 individuals treated with JAKi, as illustrated in Fig. 1. The genotypes of these 26 SNPs were analyzed in a cohort of 69 patients, and the association between patient genotypes and predictors of JAKi treatment efficacy was examined. Our prior work identified several T-cell populations, including CM Tc cells, CM Tc Fas$^+$ cells, Tc PD-1$^+$ cells, Tc CD57$^+$ cells, and Tc CD28$^+$CD27$^+$ cells, as predictors of JAKi therapeutic response in RA [16]. Specifically, lower percentages of these cells prior to treatment are associated with a better clinical outcome after 6 months of JAKi therapy [16]. Additionally, B-cell populations such as B cells, B CD21$^+$ cells, B HLADR$^+$ cells, marginal B cells, marginal B CD21$^+$ cells, marginal B HLADR$^+$ cells, SM B CD21$^+$ cells, SM B HLADR$^+$ cells, DN B cells, DN B CD21$^+$ cells, DN B HLADR$^+$ cells, B CD38dimCD21dim cells, and class non-switched memory B cells have also been found to predict JAKi efficacy [16]. Higher values of these cell populations before treatment are associated with improved outcomes after 6 months of therapy [16]. Our research results demonstrate significant associations between the following SNPs and pre-treatment immune cell percentages: rs11574891 ($p = 0.034$), rs3125003 (p = 0.034), rs61807669 ($p = 0.0073$), and Affx-4817039 ($p = 0.0027$) (Fig. 3). Furthermore, the GWAS results analysis revealed the following OR for these four SNPs: rs11574891 (OR = 3.67, 95 % CI: 2.07–6.49, $p = 10^{-5.10}$), rs3125003 (OR = 3.14, 95 % CI: 1.79–5.49, $p = 10^{-4.21}$), rs61807669 (OR = 3.17, 95 % CI: 1.86–5.40, $p = 10^{-4.31}$), and Affx-4817039 (OR = 3.20, 95 % CI: 1.84–5.62, $p = 10^{-4.31}$) (Fig. 2) (Table 2). Notably, rs11574891 and rs3125003 are located within the

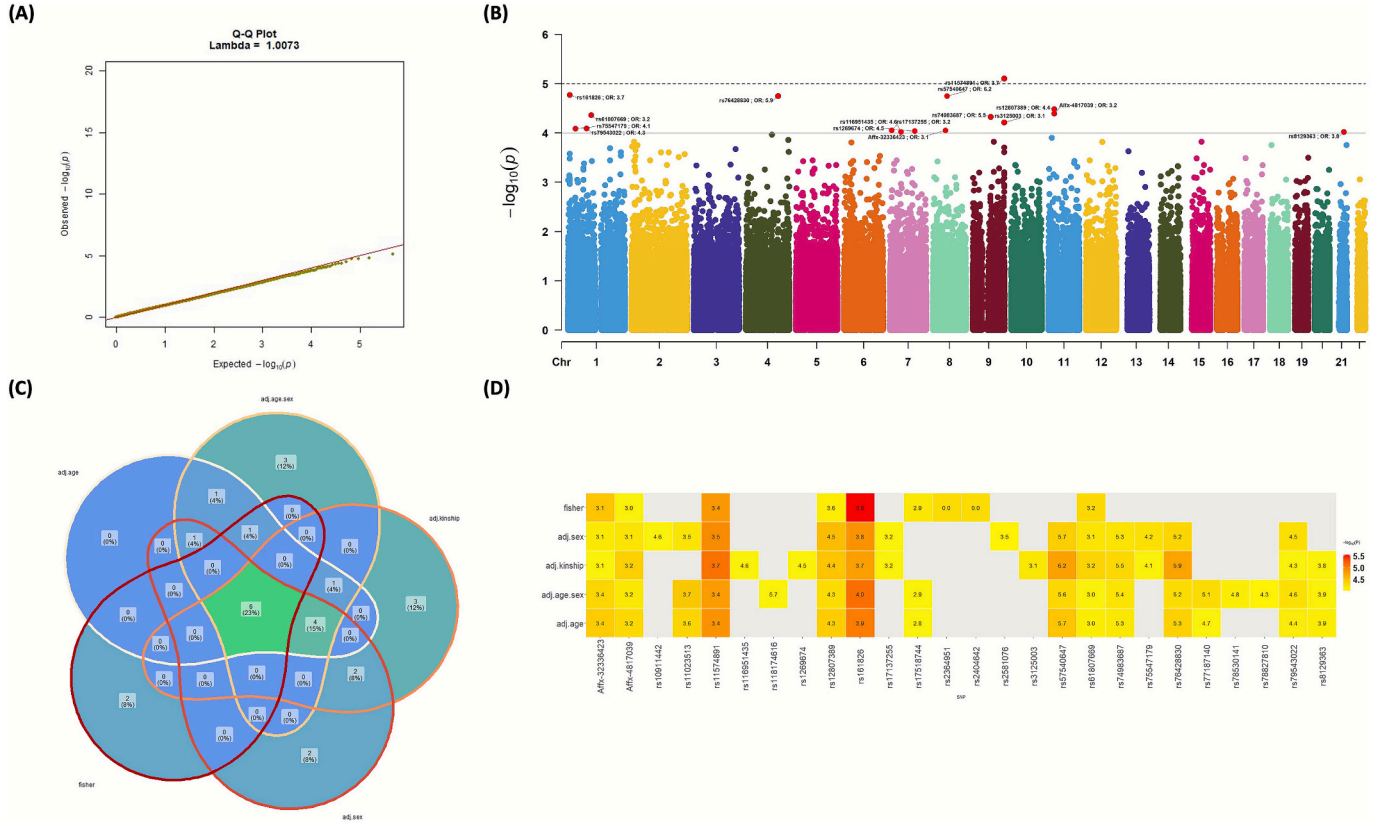


Fig. 2. GWAS of RA patients treated with JAKi and non-JAKi treatment. (A) Quantile-quantile plot showing no inflation of χ^2 test statistics ($\lambda = 1.0073$). The top SNP is highlighted. **(B)** Manhattan plot of JAKi-associated SNPs, adjusted for age and gender, with a significance threshold at $P = 1e\text{-}4$. **(C)** Venn diagram of significant SNPs from five methods, including Fisher (fisher), kinship (adj. kinship), and adjustments for age (adj. age), sex (adj. sex), or both (adj. Age.sex). **(D)** Heat map of odds ratios for 26 SNPs, indicating JAKi response in RA patients. SNPs, single nucleotide polymorphisms.

Table 2
GWAS association results for loci associated with JAKi treatment response.

SNP	Chr	Position	A1/A2[a]	MAF case	MAF control	MAF CHS	MAF global	OR (95 % CI)[b]	−log10(P)[b]	Gene
rs161826	1	7,920,278	A / G	0.75	0.44	G: 0.024	0.336	3.99 (2.18–7.30)	5.13	*TNFRSF9*
rs75547179	1	33,541,419	T / C	0.22	0.06	T: 0.028	0.014	4.19 (2.09–8.42)	4.25	*CSMD2*
rs79543022	1	82,064,628	T / C	0.20	0.06	T: 0.098	0.059	4.60 (2.19–9.65)	4.25	–
rs61807669	1	102,120,746	C / T	0.41	0.18	C: 0.079	0.238	3.17 (1.86–5.40)	4.31	–
rs10911442	1	183,819,333	C / G	0.20	0.06	C: 0.056	0.414	4.63 (2.14–10.0)	3.01	*RGL1*
rs78530141	1	230,871,637	G / A	0.17	0.05	G: 0.014	0.012	4.80 (2.21–10.4)	4.12	–
rs77187140	1	230,880,913	G / A	0.18	0.05	G: 0.014	0.012	5.06 (2.31–11.1)	4.28	*LOC124904547*
rs2581076	2	11,946,572	G / A	0.30	0.12	G: 0.190	0.146	3.47 (1.85–6.49)	4.00	–
rs2404642	4	135,158,536	T / C	0.26	0.10	T: 0.181	0.258	3.13 (1.69–5.81)	3.00	–
rs76428830	4	140,809,429	T / C	0.19	0.05	T: 0.140	0.014	5.94 (2.63–13.4)	4.75	–
rs78827810	4	185,762,624	G / A	0.19	0.05	G: 0.007	0.043	4.29 (2.06–8.93)	4.01	*SORBS2*
rs118174616	5	72,876,469	T / C	0.16	0.04	T: 0.037	0.010	5.67 (2.40–13.5)	4.09	*TNPO1*
rs116951435	7	49,123,822	A / C	0.17	0.04	A: 0.074	0.014	4.64 (2.15–10.0)	3.02	–
rs17137255	7	7,753,005	G / A	0.33	0.13	G: 0.217	0.165	3.22 (1.80–5.77)	4.05	*UMAD1*
rs1269674	7	108,280,243	G / T	0.20	0.07	G: 0.083	0.105	4.54 (2.13–9.70)	4.03	*NRCA*
Affx-32,336,423	8	54,353,568	C / A	0.67	0.40	C: 0.403	0.453	3.42 (1.90–6.14)	4.41	–
rs57540647	8	61,304,895	T / G	0.17	0.04	T: 0.042	0.055	6.17 (2.69–14.2)	4.75	*CLVS1*
rs74983687	9	77,308,025	T / C	0.17	0.05	T: 0.050	0.009	5.51 (2.42–12.5)	4.32	*VPS13A*
rs3125003	9	136,511,023	C / A	0.56	0.32	C: 0.970	0.177	3.14 (1.79–5.49)	4.21	*NOTCH1*
rs11574891	9	136,514,017	A / G	0.47	0.21	A: 0.302	0.389	3.67 (2.07–6.49)	5.10	*NOTCH1* *LOC124902310*
rs11023513	11	15,299,356	C / T	0.36	0.16	C: 0.142	0.392	3.71 (2.00–6.89)	4.46	–
rs12807389	11	25,848,127	A / G	0.28	0.10	A: 0.140	0.265	4.46 (2.21–8.97)	4.54	*LINC02699*
Affx-4,817,039	11	25,849,142	G / A	0.45	0.22	G: 0.252	0.466	3.20 (1.84–5.62)	4.31	*LINC02699*
rs2364951	17	67,575,380	G / A	0.42	0.21	G: 0.143	0.336	2.70 (1.60–4.57)	3.50	*PITPNC1* *LOC124904045*
rs17518744	18	6,979,920	G / A	0.60	0.34	G: 0.287	0.458	2.89 (1.70–4.90)	4.08	*LAMA1*
rs8129363	21	33,139,820	G / A	0.23	0.08	G: 0.108	0.036	3.70 (1.95–7.00)	3.69	–

GWAS, Genome-wide association study; JAKi, Janus kinase inhibitors; SNP, single-nucleotide polymorphism; Chr, chromosome; A, allele; MAF, minor allele frequency; CHS, Han Chinese South, China; OR, odds ratio; CI, confidence interval; P, *p* value.

[a] A1 = minor allele, A2 = major allele.

[b] The ORs, 95 % CIs and *P* values presented in the table are calculated based on data from the study adjusted for age and gender.

NOTCH1 gene, while Affx-4817039 corresponds to *LINC02699*.

Further analysis categorized rs11574891 genotypes as GA, GG, and AA, with A as the minor allele and G as the major allele. Individuals with the AA genotype exhibited significantly higher pre-treatment Naive B HLADR$^+$ cell percentages compared to those with the GA genotype (p = 0.034, Fig. 3A). As for rs3125003, genotypes were categorized as CC, TC, and TT, with C as the minor allele and T as the major allele. The CC genotype group demonstrated significantly higher pre-treatment B CD21$^+$ cell percentages compared to the TC group (p = 0.034, Fig. 3B) and significantly higher pre-treatment marginal B PD-1$^+$ cell percentages compared to the TT group (p = 0.038, Fig. 3C). Regarding rs61807669, genotypes were classified as CC, TC, and TT, with C as the minor allele and T as the major allele. The TC genotype group exhibited significantly higher pre-treatment SM B cell percentages (p = 0.0092 vs. CC, p = 0.0073 vs. TT, Fig. 3D). Additionally, the CC genotype group displayed significantly higher pre-treatment percentages of B IgM$^+$ cells (p = 0.014, Fig. 3E), class non-switched memory B cells (p = 0.025, Fig. 3F), and Naive B cells (p = 0.05, Fig. 3G) compared to the TC genotype group. For Affx-4,817,039, genotypes were categorized as AA, AG, and GG, with G as the minor allele and A as the major allele. The GG genotype group exhibited significantly higher pre-treatment percentages of Naive B cells (p = 0.014, Fig. 3H), Naive B HLADR$^+$ cells (p = 0.022, Fig. 3I), B IgM$^+$ cells (p = 0.016, Fig. 3J), DN B HLADR$^+$ cells (p = 0.047, Fig. 3K) compared to the AA genotype group. Furthermore, the AG genotype group demonstrated significantly higher pre-treatment B CD21$^+$ cell percentages compared to the AA genotype group (p = 0.0027, Fig. 3L). Conversely, the GG genotype group exhibited significantly lower pre-treatment SM B CD21$^+$ cell percentages compared to the AG genotype group (p = 0.017, Fig. 3M).

3.3. PRS models effectively differentiate JAKi patients from controls

PRS models successfully distinguished JAKi-treated patients from controls. Among the five PRS methods evaluated, the C + T model incorporating age and sex as covariates (C + T_cov) exhibited the highest classification performance, achieving an AUROC of 0.817, which was significantly superior to Lassosum (AUROC = 0.554) (Fig. 4A). Additionally, the scaled C + T_cov PRS demonstrated a more pronounced distinction between JAKi-treated and control groups compared to the standard C + T PRS (Fig. 4B). The UpSet plot demonstrated that multiple PRS models identified overlapping SNPs, indicating the presence of shared genetic markers potentially associated with JAKi response (Fig. 4C). Notably, among the 101 loci jointly identified by C + T_cov and PRSice-2 models (summarized in Supplementary Data 10), several mapped to immune-related genes, including rs11574891 and rs3125003 in *NOTCH1* and rs161826 in *TNFRSF9*. These loci showed consistent direction of effect across models. In particular, rs11574891 (OR = 3.67; 95 % CI: 2.07–6.49; $\boldsymbol{p}$ = 0.034) and rs3125003 (OR = 3.14; 95 % CI: 1.79–5.49; p = 0.034) were validated by both the C + T_cov and PRSice-2 models, while rs75547179 in *CSMD2* (OR = 4.19; 95 % CI: 2.09–8.42; $\boldsymbol{p}$ = 0.00425) was detected by four independent PRS approaches. The results suggest that these genetic variations may influence JAKi response in RA patients.

3.4. Adequate power for common variants but limited power for low-frequency SNP

Using genpwr (genpwr.calc) under a logistic model (N = 410, case rate = 0.08, two-sided α = 0.05), we evaluated power across MAF 0.07–0.34 for OR = 3 and OR = 4 under additive, dominant, recessive, and 2-df tests. A summary of the primary additive model and related results is presented in Supplementary Data 11. For common variants (MAF ≥ 0.20), power exceeded ~85–95 % at OR = 3 and ~ 95–100 % at OR = 4 (additive model), whereas the low-frequency SNP rs75547179 (MAF ≈ 0.07) showed limited power (~45–50 % at OR = 3; ~70–75 % at OR = 4).

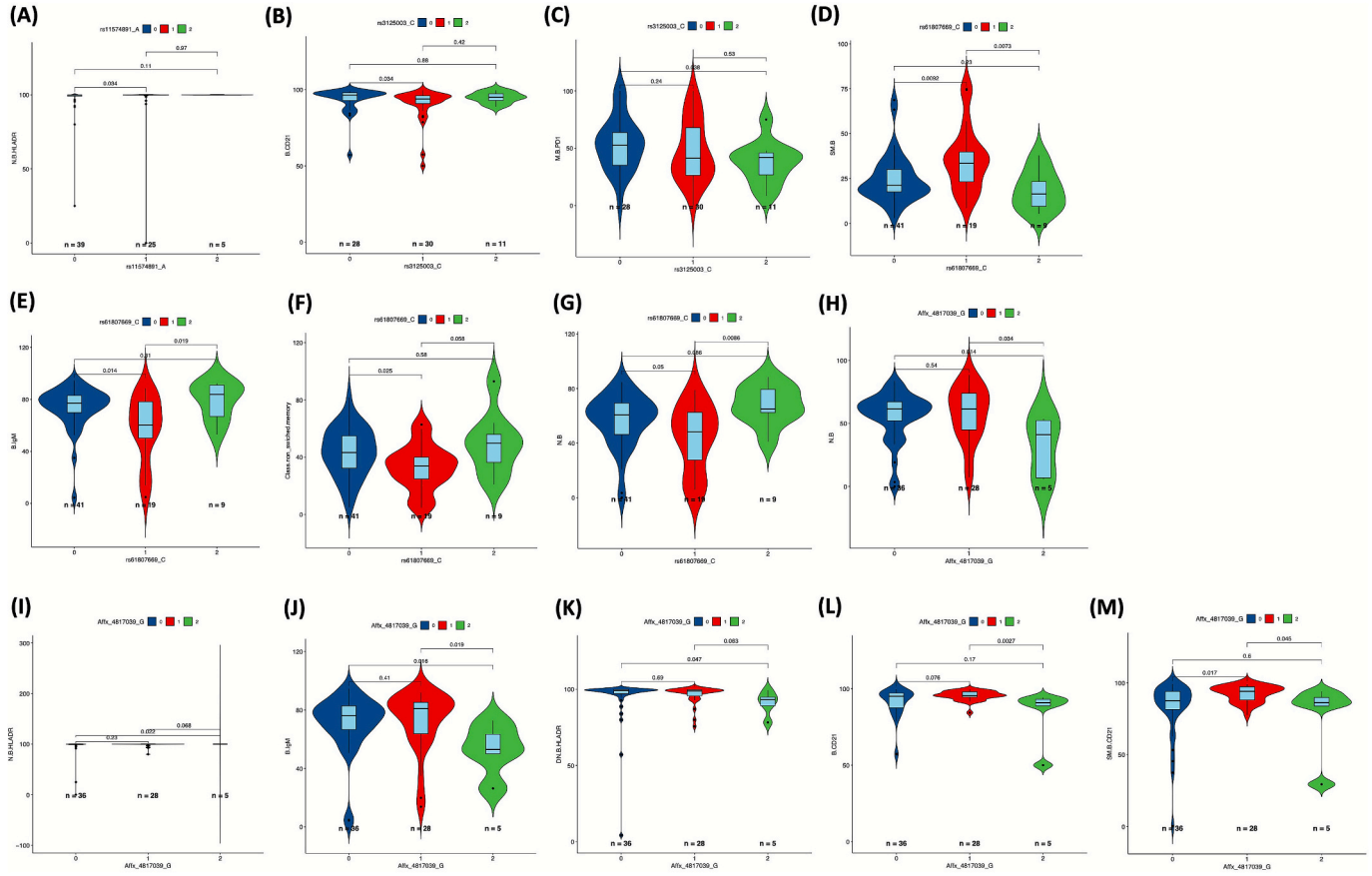


Fig. 3. Identification of SNPs correlated to different immune cells predictive of JAKi efficacy. Violin plots display the distribution, median, and interquartile range of baseline immune-cell subsets stratified by SNP genotype. '0' denotes homozygosity for the minor allele; '1' denotes heterozygosity (one copy each of minor and major allele); and '2' denotes homozygosity for the major allele. **(A)** rs11574891 (minor allele A; major allele G) is associated with Naive B HLADR$^+$ cells. **(B–C)** rs3125003 (minor allele C; major allele A) is associated with B CD21$^+$ and marginal B PD-1$^+$ cells. **(D–G)** rs61807669 (minor allele C; major allele T) is associated with switched-memory (SM) B, B IgM$^+$, class non-switched memory B, and Naive B cells. **(H–M)** Affx-4,817,039 (minor allele C; major allele A) is associated with B CD21$^+$, Naive B, SM B CD21$^+$, Naive B HLADR$^+$, B IgM, and double-negative (DN) B HLADR$^+$ cells. Statistical comparisons between genotype groups were performed using the Mann–Whitney *U* test; x-bars indicate pairwise *p*-values.

4. Discussion

To date, no GWAS has focused on predicting JAKi response in the Taiwanese RA population. Leveraging the TPMI database, this study is the first to identify 26 SNPs significantly associated with JAKi efficacy. Subsequent correlation analyses between these 26 SNPs and previously validated immunophenotypic predictors of JAKi response yielded four top candidate markers (rs11574891, rs3125003, rs61807669, and Affx-4,817,039). Furthermore, PRS modeling confirmed a substantive genetic contribution to therapeutic outcomes. To our knowledge, this is the first study to link specific SNPs to JAKi treatment response in RA. This study offers three key strengths:1) Enrollment and follow-up of 34 Taiwanese RA patients treated with JAKi for at least six months without adverse events, ensuring the integrity and reliability of clinical data; 2) Comprehensive analysis of 26 SNPs associated with JAKi efficacy, along with their relationships to immune-phenotypic markers predictive of treatment response; 3) Rigorous statistical modeling that incorporates age and sex as covariates to isolate the independent genetic contribution to JAKi treatment outcomes.

The primary objective of this study was to identify SNPs associated with clinical response to JAKi. We identified 26 SNPs significantly correlated with favorable treatment outcomes. Among them, rs161826, rs75547179, rs11574891, and rs3125003 have been previously implicated in RA pathogenesis [25–27]. Specifically, rs161826, located in *TNFRSF9* on chromosome 1, encodes the 4-1BB receptor expressed on activated T cells, promoting the proliferation of antigen-specific CD4$^+$ (Th) and CD8$^+$ (Tc) T cells and enhancing proinflammatory cytokine production [25]. rs75547179, also on chromosome 1, is located in *CSMD2*, a gene homologous to *CSMD3*, which has been reported as an RA susceptibility locus [26]. Furthermore, rs11574891 and rs3125003, both within *NOTCH1* on chromosome 9, are associated with synovial cell hyperplasia and TNF-α-induced proliferation in RA [27]. Inhibition of NOTCH1 signaling has been shown to attenuate inflammatory arthritis by modulating regulatory Treg expansion [28] (Fig. 5). Additional evidence indicates that NOTCH signaling contributes to RA pathogenesis through the regulation of angiogenesis, macrophage polarization, lymphocyte differentiation, and synovial vascular remodeling [29,30]. To the best of our knowledge, no previous study has evaluated the association between these SNPs and clinical response to JAKi. Our findings suggest that these 26 variants are significantly associated with favorable long-term JAKi response in RA patients without adverse events.

Another key aspect of this study was to identify SNPs with potential as predictive markers for JAKi clinical efficacy. Further analysis of the 26 significant SNPs revealed that only four—rs11574891 (*NOTCH1*), rs3125003 (*NOTCH1*), rs61807669, and Affx-4,817,039 (*LINC02699*)—were significantly associated with the pre-treatment proportions of specific B cell subsets predictive of JAKi response (Fig. 3). Notably, both rs11574891 and rs3125003 are located within the *NOTCH1* gene, which has been reported to play a critical role in the pathogenesis of RA [29,30]. In contrast, rs61807669 is located in a non-coding RNA region, and no studies to date have linked this variant to RA susceptibility or

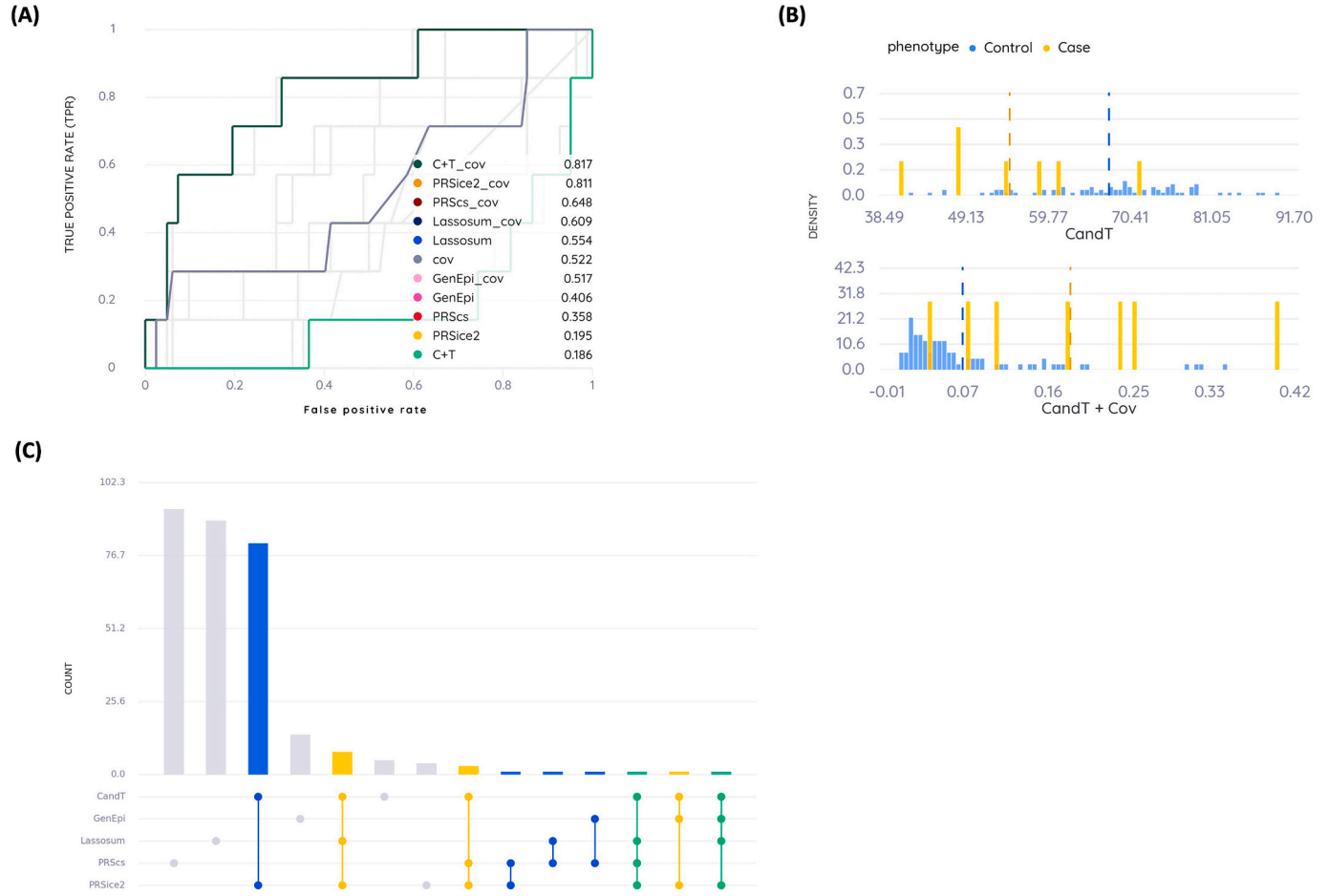


Fig. 4. Polygenic Risk Score (PRS) **of** JAKi Treatment Response **(A)** Receiver-operating-characteristic (ROC) curves of five PRS models—PRSice-2, C + T, PRScs, Lassosum, and GenEpi—evaluated with and without adjustment for age and sex ("_cov"). AUROC values are shown in the inset; the C + T_cov model achieved the highest AUROC (0.817). **(B)** Distribution of PRS in JAKi-treated cases (yellow) and controls (blue) for the unadjusted C + T (C and T) and adjusted C + T_cov (C and T + cov) models; blue and yellow dashed lines indicate the respective median PRS values. **(C)** UpSet plot showing the number of SNPs uniquely or jointly selected by the five PRS models; vertical bars indicate shared loci. Notably, *NOTCH1* variants rs11574891 and rs3125003 were consistently identified by both the C + T_cov and PRSice-2 models. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

JAKi efficacy. Interestingly, our findings demonstrated that rs61807669 was significantly correlated with the proportions of four JAKi-predictive B cell subsets: SM B cells, IgM^+ B cells, non-switched memory B cells, and Naive B cells [16], further supporting its potential predictive value. Affx-4,817,039 (also known as rs1441498) is mapped to chromosome 11 at position 25,849,142 and corresponds to the long intergenic non-coding RNA gene *LINC02699* (Table 2). Previous research has shown that vascular endothelial growth factor can upregulate *LINC02699* expression [31], but no studies have yet established a relationship between this gene and RA susceptibility or JAKi treatment response. In our study, Affx-4,817,039 was significantly associated with several immunophenotypic B cell subsets predictive of JAKi efficacy, including Naive B cells, Naive B $HLADR^+$ cells, B IgM^+ cells, DN B $HLADR^+$ cells, B $CD21^+$ cells, and SM B $CD21^+$ cells. Moreover, prior external studies have reported that immune-cell functional readouts observed during the early phase of JAKi therapy—such as monocyte STAT1 phosphorylation and neutrophil activation markers—are associated with subsequent treatment response in RA [32,33], thereby providing complementary evidence to our baseline immune-cell predictors. Taken together, these findings underscore the potential of these four SNPs as genetic biomarkers for the prediction of JAKi treatment outcomes in RA. Further validation in larger, independent cohorts is warranted to confirm their clinical applicability.

We evaluated the impact of genetic background on JAKi treatment response in Taiwanese RA patients using five PRS approaches. Age and sex were included as covariates to minimize confounding effects and enhance model robustness. Notably, four methods—PRSice-2, C + T, Lassosum, and GenEpi—consistently identified rs75547179 in *CSMD2*, a locus previously implicated in RA pathogenesis [26]. By comparing the overlap among PRS approaches, we further identified 101 SNPs shared by the two most robust models (C + T_cov and PRSice-2), as summarized in Supplementary Data 10. Several of these loci mapped to immune-related genes, including rs11574891 in *NOTCH1* and rs161826 in *TNFRSF9*, both previously reported to be associated with RA pathogenesis [25, 27, 28]. Among these, rs11574891 in *NOTCH1* also correlated with immunophenotypic predictors, suggesting a potential role in modulating JAK/STAT-dependent B-cell responses to JAKi therapy. Collectively, these findings highlight the genetic basis underlying variability in JAKi treatment response.

Despite the significant associations observed, we acknowledge that the relatively small JAKi-treated cohort ($n = 34$) constrains statistical power, particularly for low-frequency variants such as rs75547179, potentially increasing the risk of false-positive or false-negative findings. To address this concern, we performed a post-hoc power analysis, which showed adequate power (>85–95 %) for common variants but limited power (≈45–50 % for OR = 3) for low-frequency variants. This underscores the robustness of our findings for common SNPs (e.g., rs11574891, rs3125003), while highlighting the need for replication in larger, independent cohorts, particularly for low-frequency variants.

Among the 26 SNPs identified in this study, rs11574891 and

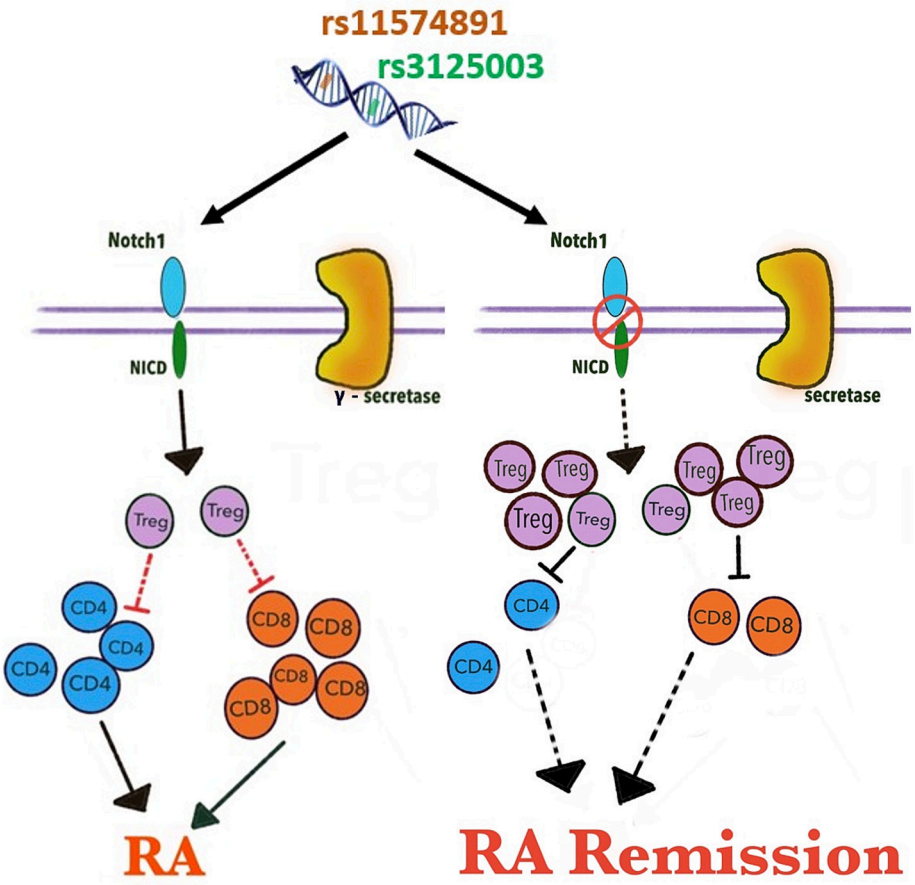


Fig. 5. Regulation of NOTCH1 signaling in rheumatoid arthritis (RA) pathogenesis and remission. This figure illustrates the role of NOTCH1 signaling in RA progression (left) and remission (right) through its effects on immune cell subsets. **(Left)** Activation of NOTCH1 signaling through γ-secretase cleavage releases NOTCH intracellular domain (NICD) reducing regulatory T cells (Tregs) and promoting $CD4^+$/$CD8^+$ T cell–mediated inflammation, contributing to RA pathogenesis. **(Right)** Inhibition of NOTCH1 increases Treg populations, restoring immune balance and suppressing inflammation, leading to RA remission.

rs3125003 were both located within the *NOTCH1* gene. Previous studies have demonstrated the involvement of *NOTCH1* in the pathogenesis of RA [28–30]. Our findings further revealed that these two SNPs were significantly associated with specific pre-treatment B-cell subsets known to predict JAKi response [16]. Notably, rs11574891 showed consistent and significant associations with JAKi efficacy across multiple PRS models, reinforcing its biological relevance as a potential predictive biomarker.

Although the aforementioned *NOTCH1* SNPs (such as rs11574891 and rs3125003) were significantly associated with JAKi efficacy, there is currently no evidence supporting their direct involvement in the JAK-STAT signaling pathway targeted by JAKi. Canonical NOTCH signaling primarily interfaces with NF-κB, MAPK, and AKT pathways rather than directly influencing the JAK-STAT axis [34]; however, prior studies have suggested that context-dependent cross-talk between the NOTCH and JAK–STAT pathways may occur in specific cell types [35]. Accordingly, the associations observed here between *NOTCH1* SNPs and JAKi efficacy are more likely attributable to immune cell–mediated indirect regulation rather than direct modulation of JAK–STAT signaling. Although both NOTCH and JAK-STAT signaling play critical roles in regulating cellular growth and differentiation, an integrated mechanism fully explaining their interaction has not yet been established. At the molecular level, downstream Hes proteins of NOTCH can bind STAT3 and recruit JAK2, thereby promoting STAT3 phosphorylation and providing direct mechanistic evidence for NOTCH–JAK/STAT cross-talk [36]. At the pathway-regulatory level, NOTCH can suppress STAT nuclear translocation, whereas JAK/STAT activity can reciprocally suppress NOTCH signaling [35]. In a cardiac burn-injury model, inhibition of *NOTCH1* reduced p-JAK2/p-STAT3 and MnSOD expression and exacerbated oxidative damage, suggesting that under stress conditions *NOTCH1* may function upstream of the JAK2/STAT3 axis [37]. Moreover, developmental and immune studies indicate that such interactions are cell-type specific and context dependent, particularly within hematopoietic and immune cells [38,39]. Nonetheless, most mechanistic evidence has been derived from non-lymphoid or developmental/oncologic models, and its clinical relevance requires further validation in RA-derived primary B/T cells.

Several studies have also demonstrated that NOTCH1 and JAK/STAT signaling play crucial roles in B- and T-cell maturation and activation. *NOTCH1* participates in marginal zone B-cell differentiation and antibody secretion and promotes plasma-cell formation [40]. In addition, *NOTCH1* contributes to peripheral T-cell activation and memory T-cell generation [41] and exerts a central role in T-cell differentiation by regulating lineage-specific gene expression [42]. Conversely, JAKi can suppress B-cell activation, proliferation, and differentiation, thereby reducing antibody production and plasma-cell proportions [43,44]. These findings provide a theoretical basis for potential convergence and functional interactions between NOTCH and JAK/STAT signaling in specific immune contexts and offer a plausible explanation for our observations. Therefore, we hypothesize that *NOTCH1* genetic variants may alter the baseline maturation status and activation thresholds of B cells, thereby indirectly modulating their dependence on JAK/STAT signaling and ultimately influencing responsiveness to JAKi therapy.

Given the pivotal role of NOTCH signaling in B- and T-cell differentiation and in setting activation thresholds in mature B cells [45], along with evidence that JAKi modulate B- and T-cell activation,

survival, and differentiation in patients with RA [46], we propose a biologically plausible yet unproven hypothesis: germline variants of *NOTCH1* (such as rs11574891 and rs3125003) may indirectly affect pharmacologic responses to JAKi by altering baseline B-cell maturation states and activation thresholds, thereby modifying their reliance on JAK/STAT signaling. Importantly, this hypothesis does not suggest that *NOTCH1* SNPs act directly on the JAK–STAT axis targeted by JAKi; rather, it posits an immune cell–mediated indirect mechanism. The observed SNP–cell-subset–efficacy associations further support this interpretation and imply that within the therapeutic context of JAKi, NOTCH–JAK/STAT interactions in B or T cells may produce a synergistic "1 + 1 > 2" effect that amplifies or modulates drug responsiveness. Although this hypothesis is primarily derived from developmental and oncologic model evidence and supported by studies describing NOTCH–JAK/STAT cross-talk within lymphoid and stromal microenvironments [35], direct validation in primary B cells from RA patients is still lacking. Therefore, this mechanistic interpretation should be regarded as biologically plausible but exploratory, pending confirmation through disease-relevant functional studies to establish its clinical and scientific implications.

This study also has several important limitations. Although our post-hoc power analysis confirmed adequate statistical power for common variants, the limited power observed for low-frequency SNPs such as rs75547179 (MAF = 0.073) suggests a higher risk of false-negative results, warranting cautious interpretation and future replication in larger, independent cohorts. Despite enrolling RA patients from multiple regions of Taiwan with comprehensive clinical data, the overall sample size was relatively modest. Furthermore, the requirement for baseline whole-blood samples for immune-phenotyping analyses related to JAKi response increased design complexity and constrained recruitment. In addition, some JAKi-treated patients had prior exposure to other biologics; although this introduces therapeutic heterogeneity, it also reflects real-world practice and enhances external validity. Importantly, rs11574891 and rs3125003 were consistently identified across multiple PRS models and were significantly associated with both baseline B-cell subsets and JAKi response, providing internal support for their potential relevance. These findings suggest the possibility of NOTCH1–JAK/STAT interplay within RA B cells. However, this study lacked an external replication cohort and direct experimental validation in RA-derived immune cells. Consequently, the proposed mechanistic explanation should be considered biologically plausible yet exploratory, requiring confirmation in RA primary immune cells and disease-relevant in vivo models, as well as replication in independent, prospectively collected datasets to strengthen its scientific and clinical significance.

In conclusion, this study represents the first GWAS leveraging TPMI data to investigate SNPs associated with treatment efficacy under stable JAKi therapy in Taiwanese patients with RA. Future integration of these genetic variants with clinical parameters and laboratory biomarkers may enhance the precision of therapeutic-response prediction and advance personalized medicine for RA. However, validation in larger-scale cohorts remains necessary before clinical application.

Authors contribution

SWL was responsible for Methodology development, Conceptualization, Investigation, Data curation, Analysis, Validation, Project administration, Resource management, Visualization, as well as drafting the Original Manuscript and participating in the Review & Editing process. JFL, CGH and TYH provided assistance in the analysis of GWAS data. YTC and FCL were involved in Methodology refinement, Data curation, Validation, Conceptualization, Project administration, and contributed to Review writing and Supervision. All authors critically reviewed and revised the manuscript and ultimately approved the final version for publication.

CRediT authorship contribution statement

Shan-Wen Lui: Writing – review & editing, Writing – original draft, Visualization, Validation, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Feng-Cheng Liu:** Writing – original draft, Visualization, Validation, Supervision, Project administration, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Jun-Fu Lin:** Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Chia-Jung Hsu:** Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Formal analysis, Data curation, Conceptualization. **Ting-Yu Hsieh:** Writing – original draft, Visualization, Validation, Formal analysis, Data curation, Conceptualization. **Yu-Tien Chang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.intimp.2025.115825.

Data availability

Data are available from the corresponding author upon reasonable request.

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